Complexity and approximation exercises

A11-R01 Encoding and pseudopolynomial time

An algorithm tests 0 through t in t+1 constant-time steps, where binary t=2^m. Express its time in t and in bit length N. What changes under unary encoding? Explain why a graph instance must include the encoding length of k.

A11-R02 A quantifier gap in a verifier

A validator checks that a certificate lists k distinct valid vertices but does not check edges. A second version checks only consecutive pairs $c_1c_2, c_2c_3, \ldots$. Give a no instance and an accepted certificate for each. State NP completeness and soundness with quantifiers.

A11-R03 Build a verifier from a decider

Prove $P \subseteq NP$ by specifying the certificate, verifier, length bound, and both correctness directions. Does this prove $P \neq NP$ or $NP \subseteq P$?

A11-R04 Audit reduction directions

Let A be NP-complete and $B \in NP$. Which condition proves B NP-complete, and why? (a) $A \leq_p B$. (b) $B \leq_p A$. (c) $A \leq_p B$ and B has an exponential-time algorithm. What follows from $A \leq_p B$ and $B \in P$?

A11-R05 Prove composition time

f reduces A to B in $O(N^2)$ time. g reduces B to C in $O(M^3)$ time on length M. Define h, prove its semantics, and bound its time. Why does making N calls to a polynomial-time routine alone not guarantee polynomial total time?

A11-R06 Build the full 3SAT→CLIQUE graph

For $F=(x \vee \neg y \vee z) \wedge (\neg x \vee y \vee z) \wedge (x \vee y \vee \neg z)$, use the lecture numbering to list all edges, the vertex/edge counts, and k. Give a satisfying assignment and clique. Explain why 4 and 7 are not adjacent but 6 and 7 are. Reconstruct both correctness directions.

A11-R07 Repair constructions with small counterexamples

Give an unsatisfiable formula whose graph has an m-clique under each bug: (a) allowing consistent edges within a clause; (b) connecting complementary literals across clauses. Repeated literals are allowed, with three occurrences per clause. Explain the empty-conjunction and tautological-clause boundaries.

A11-R08 Recover a maximum clique from decision

Assume a polynomial-time CLIQUE decider D. Recover ω(G) and output a maximum clique. Bound the number of calls, prove correctness and polynomial time, and explain why this is not a single many-one reduction.

A11-Z01 Circuitizing a verifier

A fixed verifier has $p(n) = n$, $T(n) = n^2 + 1$, and $s(n) = 3n^2 + 5$. Give asymptotic bounds for layers, free witness inputs, and gate count. Explain why hardwiring x still represents witness existence.

A11-Z02 Sampling success

Suppose 100 clauses each contain three distinct variables. Lower-bound the chance that one random assignment satisfies at least 75 clauses. Upper-bound the chance that 10 independent trials all fail to reach 75. Why is independence between clauses unnecessary?

A11-Z03 Nondeterminism and verifiers

Prove the equivalence of two NP definitions: a nondeterministic Turing machine runs at most $T(n)$ steps on every branch, where $T(n)$ is a fixed polynomial, and at least one branch accepts; and a deterministic polynomial-time verifier has a polynomial-length witness. Explain how to encode branching choices, handle shorter branches and invalid encodings, and prove both directions.

A11-X01 Gate encoding

Encode z = a AND b in CNF. For a=1, b=0, determine z and check all clauses. Explain why the clauses reject the other z value.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

A11-X02 Complement and cover

Let G have vertices 1,2,3,4 and edges 12,23,34. Set k=2. Construct the complement, find a 2-clique of G, and the corresponding vertex cover of the complement. Prove the general equivalence.

A11-X03 Hamiltonian cycle to TSP

For the four-cycle 12,23,34,41, form the complete weighted graph of the reduction. Give a tour at the threshold. Then remove edge 41 from the source and decide whether any target tour reaches the threshold. Justify without listing all tours.

A11-X04 Decimal encoding

For $F=(x_1 \vee x_2 \vee x_3) \wedge (\neg x_1 \vee x_2 \vee \neg x_3)$, construct the target and the numbers for $x_1=x_2=$true, $x_3=$false, including slack. Show their sum and explain why an arbitrary solution determines a satisfying assignment.

A11-X05 Edge gadget and cover

For the 12-vertex edge gadget, take x=0 and y=1 in the middle-edge notation of the slides. List the full path, identify its external ports, and explain why the x=y=0 case cannot occur in a Hamiltonian cycle of the constructed graph. What source cover does the 25-vertex path example encode?

A11-A01 Directions and values

For minimization A=18, OPT=12 and maximization A=12, OPT=18, compute ratios and guarantees. Why is $A \leq \rho OPT$ insufficient for maximization?

A11-A02 Cover rounds and a tight bound

For E={ab,bc,cd,de,ef}, select ab, cd, ef. List residual edges, C, and a lower bound. Prove the ratio is exactly 2.

A11-A03 Maximal versus maximum

On a-b-c-d select bc. Prove M is maximal but not maximum, and refute taking one arbitrary endpoint per matching edge.

A11-A04 From walk to tour

For the unit square and MST={AB,BC,CD}, list the full walk and shortcut tour, compute costs, prove OPT, and identify the assumptions in the factor-2 proof.

A11-A05 A nonmetric counterexample

Perimeter edges cost 1 and AC=BD=M>4. Use MST AB,AD,BC rooted at A, visiting D first. Prove the preorder ratio is unbounded.

A11-A06 The full set-cover charging proof

Use A={1,2,3,4}, B={1,2,5}, C={3,4,6}, with alphabetical ties. Find greedy, OPT, and charges. Prove the local harmonic bound and global H_d guarantee.

A11-A07 Trace every trimmed list

For x=(104,102,201,101), t=308, ε=0.4, show merged and trimmed lists, deletion reasons, returned subset, OPT, and actual ratio.

A11-A08 Two obligations of an FPTAS proof

Prove the error invariant and accumulated product bound. Prove polynomial bit complexity in input length and 1/ε. Explain why O(nt) is insufficient.

A11-Y01 Weighted set cover

$U = \{1, 2, 3, 4\}$; $A = U$ costs 8, $B = \{1, 2\}$ costs 2, and $C = \{3, 4\}$ costs 2. Run cost-per-new-element greedy, breaking ties alphabetically. Give the choices, element charges, and OPT. If A costs $M > 4$, what ratio results from choosing the most new elements?

A11-Y02 Conditional expectation

For $(a \vee b \vee c)$ and $(\neg a \vee b \vee c)$, compute the initial expectation and the conditional expectations for $a = 0$ and $a = 1$. After choosing $a = 0$, compare $b = 0$ and $b = 1$, and give an assignment satisfying both clauses.

A11-Y03 Weighted vertex cover

A path $a - b - c$ has weights 2, 3, 2. Prove the LP optimum is 3, round $(0, 1, 0)$, and find the integer optimum.

A11-Y04 Randomized set cover

$U = \{1, 2\}$ has six unit-cost candidate sets: three contain only 1 and three contain only 2. Use LP values $x = 1/3$ for all sets. Compute LP^*, k, selection probabilities, the output, and its cost. Why does this example not prove coverage in every instance?

A11-Y05 Nonmetric TSP

Use path $1 - 2 - 3 - 4$ as a Hamiltonian-cycle input with $R = 2$. Give the nonedge cost and threshold. Enumerate the three tours starting at 1 up to reversal and their costs; explain the gap decision.

A11-C01 Three-leaf star metric

Four vertices o, a, b, c have center-to-leaf distance 1 and leaf-to-leaf distance 2. Find the odd vertices of the star MST, a minimum perfect matching and its cost. Give an Euler closed walk, its shortcut tour and cost, and OPT. Why must a parallel edge be retained?

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd edition, chapters 34 and 35. MIT 6.046J Fall 2004, Lecture 24, approximation algorithms. Original handwritten algorithms notes, complexity and approximation pages.

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